01 / PROBLEM AND CONTRIBUTION

# Battery-powered LED torch PCB

Independent hardware design · KiCad | PCB design study

**DESIGN AT A GLANCE**

Created a compact LED torch board with a battery holder, switch, resistor, and LED, and included Gerber and drill outputs for the design.

## Engineering question

How can a simple LED circuit become a compact, mechanically coherent PCB?

## Situation

This introductory hardware project uses a direct battery-switch-resistor-LED circuit. Its board outline and component arrangement follow the battery holder and accessible switch.

## Task

Capture the circuit, assign footprints, route the compact board, and prepare manufacturing-output files.

## My contribution

Schematic capture, PCB implementation, and design documentation in the linked personal GitHub project.

## Tools and concepts

KiCad / Schematic capture / PCB layout / LED circuits / Gerber outputs

02 / IMPLEMENTATION

# Circuit to board

### 01 / Design step

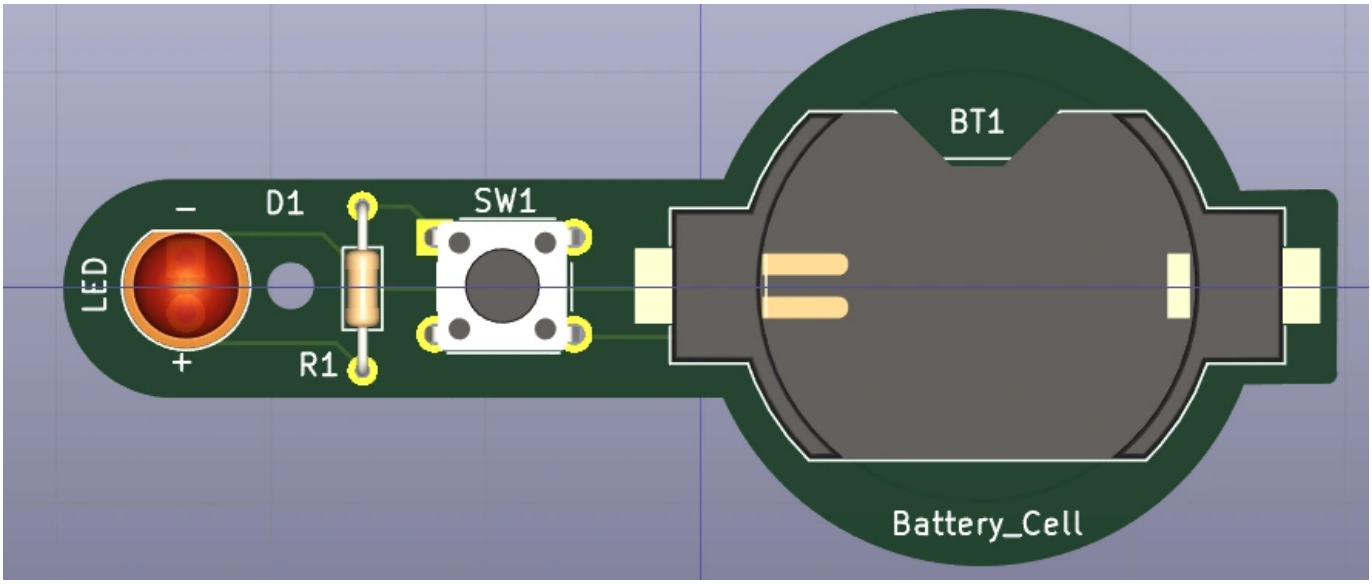
Captured the battery, switch, resistor, and LED path in KiCad. The source schematic uses a resistor-labelled R1 rather than a specified current-setting value.

### 02 / Design step

Arranged the battery holder, LED, resistor, and switch within the shaped board. The file contains two copper layers, five footprints including a mounting feature, 14 track segments, and no vias.

### 03 / Design step

Provided front/back copper, silkscreen, mask, edge-cut, and drill exports in the Gerber directory, alongside a board render and schematic preview.




**LED torch board render**

Repository 3D render showing the battery holder, LED, switch, and resistor.

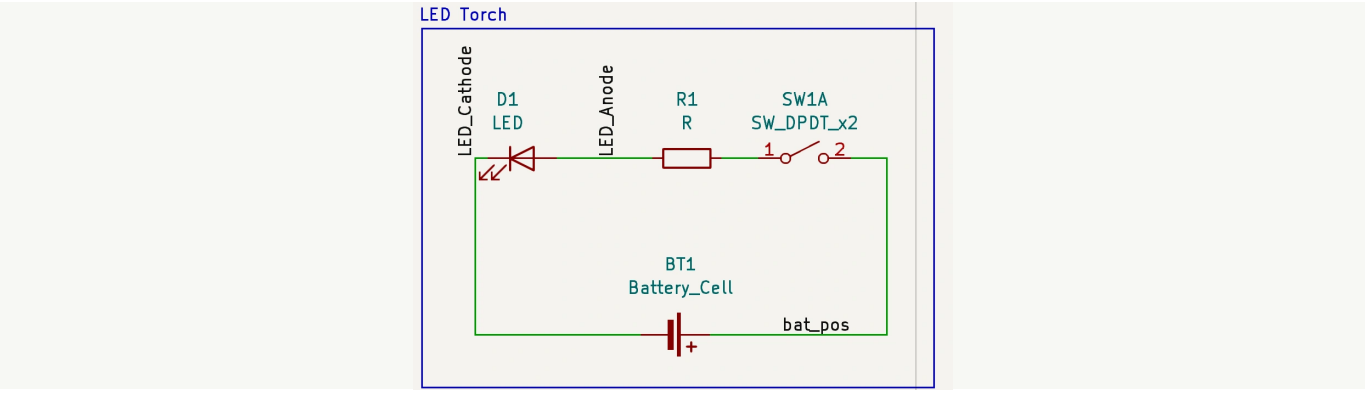
03 / EVIDENCE AND SCOPE

# Design evidence and next steps

| | |
|---|---|
| **Board file** | 2 copper layers |
| **Routing vias** | 0 |
| **Available outputs** | Gerber + drill |

## Available evidence

The repository includes the editable circuit and board, a 3D preview, and manufacturing-output files. The schematic supports a direct resistive LED circuit, not the generic transistor/boost alternatives described in the README.




**LED torch schematic**

Repository schematic of the direct battery-powered LED circuit.

## Scope and limitations

R1 has no numeric value in the checked-in schematic. LED current, brightness, battery runtime, fabrication, and bench operation are therefore not claimed. Gerber availability does not by itself establish manufacturing readiness.

### Next steps - proposed work

- Choose and document R1 from the intended battery and LED operating point.
- Run ERC/DRC and document current, brightness, and battery-runtime measurements.

Source: PCB-design-projects GitHub repository: LED Torch; README, KiCad schematic/board, and repository previews. Reviewed 11 November 2026.

Open the project repository